

Instructions.

1. **Due March 13, 11:59 PM uploaded to Blackboard as a single pdf and a single ZIP archive. This deadline is firm. Late uploads will not be accepted.**
2. This project will be done in a team of two members. You may **NOT** discuss or consult with any students outside your team.
3. All work must be typeset in L^AT_EX. Handwritten or scanned submissions will not be accepted.
4. All figures must have axis labels and descriptive captions. Tables, if any, must have descriptive captions.
5. This document may be updated if a mistake is found or further clarification is needed. A Blackboard announcement will be posted as soon as this document is updated.
6. Submission guidelines.
 - (a) Label your report as ENME408_S26_Project01_<lastnames>.pdf.
 - (b) Include all figures and tables with captions in your report.
 - (c) You may include MATLAB scripts and representative outputs as Appendices. Label the appendices and cite them in your report appropriately.
 - (d) Place all TEX files, MATLAB scripts, and Simulink models in a folder, then archive it. Label your archive as ENME408_S26_Project01_<lastnames>.zip.
 - (e) Upload your pdf file and the archive folder to Blackboard before the deadline.
7. Honor code: “I have neither received nor given aid to anyone outside my team during this project.”

Sign/print your names here.

If you cannot sign/print in the box above, please write the honor code on the first page of your submission and sign it.

8. Reports with substantial similarity will be reported to the UMBC Academic Conduct Committee.
9. Include a **credit author statement** in your report. The statement recognizes individual author contributions and facilitates collaboration. For example,
Author 1: Methodology, Software, Visualization
Author 2: Data curation, Writing, Original draft preparation, Validation.

1. Derive the multicopter equations of motion (EOM).

- Frames and angles.** Define the inertial frame F_A (use NED) and body frame F_D , and the intermediate frames. Write the orientation matrices that relate the various frames.
- Position, velocity, and acceleration.** Let c denote the center of mass of the multicopter and let w be a fixed point fixed on Earth. Let $\vec{r}_{c/w} \triangleq p_1 \hat{i}_A + p_2 \hat{j}_A + p_3 \hat{k}_A$. Write the expressions for the physical velocity vector $\vec{v}_{c/w/A}$ and the physical acceleration vector $\vec{a}_{c/w/A}$ in terms of p_1, p_2, p_3 , their derivatives, and the inertial frame F_A axes.
- Equations of motion.** Let the gravity vector be given by $\vec{g} = g \hat{k}_A$ and consider a multicopter with mass m and inertia matrix $\vec{J}_{B/c} \Big|_{D} = J = \text{diag}(J_1, J_2, J_3)$ that can exert a thrust $\vec{T} = -T \hat{k}_D$ applied at c , where $T \geq 0$, and a torque $\vec{\tau} = \tau_1 \hat{i}_D + \tau_2 \hat{j}_D + \tau_3 \hat{k}_D$, where $\tau_1, \tau_2, \tau_3 \in \mathbb{R}$. Using Newton's 2nd law, Poisson's equation, and Euler's equation, write the equations of motion of the multicopter. In particular, obtain expressions for $[\ddot{p}_1 \ \ddot{p}_2 \ \ddot{p}_3]^T$, $[\dot{\psi} \ \dot{\theta} \ \dot{\phi}]^T$, and $[\dot{\omega}_1 \ \dot{\omega}_2 \ \dot{\omega}_3]^T$, where ψ, θ, ϕ are the 3-2-1 Euler angles and $\omega_1, \omega_2, \omega_3$ are the body fixed angular velocities.
- State-space form.** Write the equations of motion in the state-space form $\dot{x} = f(x, u)$, where

$$x \triangleq [p_1 \ p_2 \ p_3 \ \dot{p}_1 \ \dot{p}_2 \ \dot{p}_3 \ \psi \ \theta \ \phi \ \dot{\omega}_1 \ \dot{\omega}_2 \ \dot{\omega}_3]^T, \quad u \triangleq [T \ \tau_1 \ \tau_2 \ \tau_3]^T. \quad (1)$$

2. Simulate the EOM in Simulink.

- Model structure.** Implement the equations of motion using vectors and matrices to compute $\dot{x}$. Use the integrator blocks to obtain the state x for $t \in [0, 10]$ s in the case where $x(0) = 0$ and $u(t) = [mg + \sin(\pi t) \ 0 \ 0 \ 0]^T$. Plot the position, velocity, and the Euler angles versus time.
Hint: Use a fixed-step solver (e.g., RK4) with a small step size (e.g., $\Delta t = 10^{-4}$ s).

3. Open-loop control.

- Let $m = 1$ and $J = \text{diag}(0.01, 0.01, 0.02)$. Consider the desired trajectory

$$p_1(t) = \cos(\Omega t), \quad p_2(t) = \sin(\Omega t), \quad p_3(t) = 0, \quad (2)$$

with $\Omega = 0.1 \times 2\pi$ rad/s. Using the translational equations of motion, compute the open-loop force vector $\vec{f} = f_1 \hat{i}_A + f_2 \hat{j}_A + f_3 \hat{k}_A$ required for the quadrotor to track this circular trajectory. Plot $p_1(t)$, $p_2(t)$, and $p_3(t)$ over time.

- Use `plot3(x,y,z)` to show the position path in NED coordinates.
- Vector overlay.** Use `quiver3` to overlay velocity vectors along the trajectory.
 - Downsample the vectors (e.g., every 10–50 samples) so the plot stays readable.
 - Apply a scaling factor so arrows are visible but not overwhelming.

Next, plot f_1, f_2 , and f_3 versus time in a single plot.

Hint: Recall that $\vec{f} = m \vec{a}_{c/w/A}$.

- Assume $\psi(t) = 0$. Compute the roll and pitch angles $\phi(t)$ and $\theta(t)$ required to realize the desired force vector $f(t)$. Plot $\phi(t)$ and $\theta(t)$ as functions of time.
Hint: First, obtain an expression for T . Then, use Newton's Second Law to obtain $\phi(t)$ and $\theta(t)$.
- Using the Poisson's equation, compute the body-fixed angular velocity vector $\omega_{D/A|D}(t)$ corresponding to the Euler angle trajectories. Plot the components of $\omega_{D/A|D}(t)$ over time.
- Using Euler's rotational equations of motion, compute the control torque $\tau(t)$ required to generate the angular velocity $\omega_{D/A|D}(t)$. Plot the components of $\tau(t)$ over time.
- Simulate the full quadrotor dynamics under the computed open-loop thrust T and torques τ_1, τ_2, τ_3 for $t \in [0, 10]$ s and confirm whether the resulting trajectory tracks the desired circular reference. Use appropriate initial conditions.

4. Effect of perturbations.

- (a) Add a small random noise to the initial position, that is,

$$p_1(0) = 1 + w_1, \quad p_2(0) = w_2, \quad p_3(0) = w_3, \quad (3)$$

where w_i are random numbers generated in Matlab using `randn`. With the perturbed initial position, simulate the full quadrotor dynamics using the previously computed open-loop forces and torques for $t \in [0, 10]$ s. Comment on the effect of initial position perturbations on the position deviations from the desired circular trajectory.

- (b) Add a small random noise to the initial velocities, that is,

$$\dot{p}_1(0) = w_1, \quad \dot{p}_2(0) = \Omega + w_2, \quad \dot{p}_3(0) = w_3, \quad (4)$$

where w_i are random numbers generated in Matlab using `randn`. With the perturbed initial velocity, simulate the full quadrotor dynamics using the previously computed open-loop forces and torques for $t \in [0, 10]$ s. Comment on the effect of initial velocity perturbations on the position deviations from the desired circular trajectory.